AI Audiometry Research: Feasibility Study

AudioApp AI-Powered Hearing Assessment

Date: January 16, 2026
Status: Research Complete
Purpose: Evaluate feasibility of replacing audiologists with AI for hearing assessment interpretation

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Executive Summary

This document provides comprehensive research on using AI/ML models to automate audiometric interpretation, potentially replacing the need for audiologists in certain workflows. Based on current research and FDA precedents, **this is technically feasible** with certain limitations.

Key Findings

Aspect	Feasibility	Notes
Audiogram Classification	High	97.5% accuracy achieved with deep learning
Air-Bone Gap Detection	High	86% accuracy in predicting middle-ear problems
Hearing Loss Type Diagnosis	High	Conductive, sensorineural, mixed - well studied
Severity Classification	High	Based on standard dB HL thresholds
Pattern Recognition	High	Sloping, flat, cookie-bite, notched patterns
FDA Approval Precedent	Exists	Apple AirPods Pro HAF, Tuned AI cleared in 2024-2025
Full Audiologist Replacement	⚠ Medium	Limited to routine interpretation; complex cases need specialists

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Part 1: What Audiologists See and Determine

1.1 The Complete Audiological Test Battery

A professional audiologist can perform and interpret the following tests:

```
┌──────────────────────────────────────────────────────────────────────
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│                AUDIOLOGICAL TEST BATTERY                  │
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│                                                  │
│
┌──────────────────────────────────────────────────────────────────────
────────────────────────────────────┐    │
│   │            PURE TONE AUDIOMETRY (PTA)              │    │
│   │   • Air Conduction (AC) - Entire auditory pathway          │    │
│   │   • Bone Conduction (BC) - Inner ear only              │    │
│   │   • Frequencies: 250, 500, 1000, 2000, 3000, 4000, 6000, 8000 Hz  │
│   │   • Output: Audiogram with thresholds in dB HL             │    │
│
```

```
│ └──────────────────────────────────────────────────────────────────────
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│                 ↓                 │
│
┌──────────────────────────────────────────────────────────────────────
──────────────────────┐ │
│ │          SPEECH AUDIOMETRY          │ │
│ │ • Speech Reception Threshold (SRT) - 50% word recognition │ │
│ │ • Word Recognition Score (WRS) - % of words understood │ │
│ │ • Speech Discrimination Score (SDS)          │ │
│ │ • Most Comfortable Level (MCL)          │ │
│ │ • Uncomfortable Loudness Level (UCL)          │ │
│
└──────────────────────────────────────────────────────────────────────
──────────────────────┘ │
│                 ↓                 │
│
┌──────────────────────────────────────────────────────────────────────
──────────────────────┐ │
│ │          IMMITTANCE AUDIOMETRY          │ │
│ │ • Tympanometry - Middle ear function          │ │
│ │ • Acoustic Reflexes - Stapedius muscle response          │ │
│
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┌──────────────────────────────────────────────────────────────────────
──────────────────────┐ │
│ │          OBJECTIVE TESTS (No patient response needed) │ │
│ │ • OAE (Otoacoustic Emissions) - Cochlear outer hair cells │ │
│ │ • ABR (Auditory Brainstem Response) - Neural pathway │ │
│
└──────────────────────────────────────────────────────────────────────
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────────────────────────────────┘
```

1.2 Air Conduction vs Bone Conduction: The Critical Relationship

How They Work

Test	What It Measures	Pathway	Equipment
Air Conduction (AC)	Complete auditory system	Outer ear → Middle ear → Inner ear → Auditory nerve	Headphones (TDH-39, insert earphones)
Bone Conduction (BC)	Inner ear function only	Vibrator → Skull → Inner ear → Auditory nerve	Bone vibrator on mastoid

The Air-Bone Gap (ABG)

```
        Normal Hearing      Conductive Loss     Sensorineural Loss     Mixed Loss
       ___________________________________    _________________________________
______________________________   ______________________________
dB HL     AC    BC          AC    BC          AC    BC          AC    BC
 0 ────────  ●════════●          ●            ●            ●
10 ────────            ↓         ↓            ↓
20 ────────            ↓   ‖       ↓   ‖        ↓   ‖
30 ────────            ↓   ↓     ↓   ↓        ↓   ↓
40 ────────            ●═══════╝          ●═══════●          ●═══════╝
50 ────────            ↑                      ↑
                 └──── Air-Bone Gap          └──── Air-Bone Gap
                    (>10-15 dB)           with BC shift
```

Key Diagnostic Rule:

- **No Air-Bone Gap** (AC ≈ BC): Sensorineural hearing loss (inner ear/nerve damage)
- **Significant Air-Bone Gap** (AC > BC by 10-15+ dB): Conductive component (outer/middle ear problem)
- **Both Elevated with Gap**: Mixed hearing loss (both inner ear and outer/middle ear problems)

1.3 Audiogram Symbol Standards

Symbol	Meaning	Color
O	Right Ear Air Conduction	Red
X	Left Ear Air Conduction	Blue
<	Right Ear Bone Conduction (unmasked)	Red
>	Left Ear Bone Conduction (unmasked)	Blue
[	Right Ear Bone Conduction (masked)	Red
]	Left Ear Bone Conduction (masked)	Blue
↓	No Response (at maximum output)	-

1.4 Hearing Loss Classification by Severity

Degree	PTA Range (dB HL)	Functional Impact
Normal	-10 to 25 dB	No significant difficulty
Mild	26 to 40 dB	Difficulty with soft speech, distant sounds
Moderate	41 to 55 dB	Difficulty with normal conversation
Moderately-Severe	56 to 70 dB	Difficulty with loud speech
Severe	71 to 90 dB	Can only hear very loud sounds
Profound	>90 dB	Hearing aids usually insufficient

1.5 Audiogram Configuration Patterns

Pattern Types and Clinical Significance

Configuration	Pattern Description	Common Causes	Visual Pattern
Flat	Equal loss across frequencies	Genetic, otosclerosis, some medications	
Sloping	Progressive loss at higher frequencies	Presbycusis (age-related), noise exposure	

Rising	Better high-frequency hearing	Ménière's disease, genetic	/////////
Cookie-Bite	Mid-frequency dip	Genetic (often hereditary)	__/
Notched	Sharp dip at 3-6 kHz	Noise-induced hearing loss (NIHL)	▬_/▬
Precipitous	Steep drop at high frequencies	Sudden hearing loss, ototoxicity	▬\\ \\

1.6 What an AI Model Must Determine

For complete audiological assessment, the AI must output:

```
class AudiologicalAssessment {
  // 1. Type of Hearing Loss
  HearingLossType type;  // normal, conductive, sensorineural, mixed

  // 2. Degree of Hearing Loss (per ear)
  HearingLossDegree rightEarDegree;  // normal, mild, moderate, moderately-severe, severe, profound
  HearingLossDegree leftEarDegree;

  // 3. Configuration
  AudiogramConfiguration configuration;  // flat, sloping, rising, cookie-bite, notched, precipitous

  // 4. Symmetry
  bool isSymmetrical;  // Are both ears similar?

  // 5. Pure Tone Averages
  double ptaRightEar;  // Average of 500, 1000, 2000, 4000 Hz
  double ptaLeftEar;
  double speechFrequencyAverage;  // Average of 500, 1000, 2000 Hz

  // 6. Air-Bone Gaps
  Map<int, double> rightEarABG;  // Per frequency: AC - BC
  Map<int, double> leftEarABG;
  bool hasSignificantConduction;  // ABG > 10-15 dB

  // 7. Speech Audiometry Correlation
  bool srtCorrelatesWithPTA;  // SRT should be within 5-12 dB of PTA
  double wordRecognitionScore;

  // 8. Clinical Flags
  List<ClinicalFlag> flags;  // Asymmetry, SRT-PTA mismatch, notch pattern, etc.

  // 9. Recommendations
  List<Recommendation> recommendations;  // Refer to ENT, hearing aid, etc.
}
```

Part 2: AI/ML Models for Audiometry

2.1 Proven AI Architectures

Research has demonstrated high accuracy with several model types:

Model Type	Accuracy	Use Case	Notable Research
ResNet-101	97.5%	Audiogram image classification	AutoAudio model
Convolutional Neural Networks (CNN)	90-97%	Image-based audiogram analysis	Multiple studies
Recurrent Neural Networks (RNN)	93%+	Sequential threshold data	Time-series analysis
LSTM	90%+	Temporal patterns in audiometry	Complex pattern recognition
Random Forest	85-90%	Feature-based classification	Traditional ML approach
Support Vector Machines (SVM)	80-90%	Binary/multi-class classification	Baseline models
Logistic Regression	75-85%	Simple classification tasks	Interpretable results

2.2 Two AI Approaches

Approach A: Image-Based Classification

```
┌─────────────────────────────────────────────────────┐
│              Image-Based Pipeline                    │
├─────────────────────────────────────────────────────┤
│                                                      │
│   Audiogram Image (JPG/PNG)                          │
│         ↓                                            │
│   Preprocessing (crop, normalize, resize to 224×224) │
│         ↓                                            │
│   CNN (ResNet-101 / VGG-16 / EfficientNet)           │
│         ↓                                            │
│   Feature Extraction                                 │
│         ↓                                            │
│   Classification Head                                │
│         ↓                                            │
│   Output: [Normal | Conductive | Sensorineural | Mixed] │
│                                                      │
└─────────────────────────────────────────────────────┘

Pros:
    Works with existing audiogram images
    High accuracy (97.5% reported)
    Can process varied visual formats

Cons:
    Requires image preprocessing
    Sensitive to audiogram visual variations
    More computationally intensive
```

Approach B: Numerical Data Classification

```
┌─────────────────────────────────────────────────────────────────┐
│              Numerical Data Pipeline                  │
├─────────────────────────────────────────────────────────────────┤
│                                             │
│   Threshold Data (16 features):                       │
│   [AC_R_250, AC_R_500, ..., BC_R_4000, BC_L_4000]       │
│        ↓                                     │
│   Feature Engineering:                            │
│   - PTA calculations                              │
│   - Air-Bone Gap calculations                        │
│   - Slope calculations                            │
│   - Pattern detection features                       │
│        ↓                                     │
│   ML Model (Random Forest / XGBoost / Neural Network)    │
│        ↓                                     │
│   Multi-output Classification:                      │
│   - Type: [Normal | Conductive | SNHL | Mixed]          │
│   - Degree: [Normal | Mild | Moderate | ...]            │
│   - Config: [Flat | Sloping | Notched | ...]            │
│                                             │
└─────────────────────────────────────────────────────────────────┘

Pros:
    Faster inference
    More interpretable
    Works directly with app-generated data
    Easier to add rules/heuristics

Cons:
    Requires clean numerical input
    May miss visual patterns
```

2.3 Recommended Architecture for AudioApp

For a mobile app generating its own data, **Approach B (Numerical)** is recommended:

```
class AIAudiologist {
  // Input Features (16 core + derived features)
  final List<int> frequencies = [250, 500, 1000, 2000, 3000, 4000, 6000, 8000];

  // Core Feature Vector (minimum 16 values)
  // AC_Right: 8 values (one per frequency)
  // AC_Left: 8 values
  // BC_Right: 6 values (250, 500, 1000, 2000, 3000, 4000 Hz)
  // BC_Left: 6 values
  // Total: 28 threshold values

  // Derived Features
  double calculatePTA(List<double> thresholds) {
    // (500 + 1000 + 2000 + 4000) / 4
  }

  List<double> calculateAirBoneGaps(List<double> ac, List<double> bc) {
    // AC - BC at each frequency
  }

  double calculateSlope(List<double> thresholds) {
    // Linear regression slope across frequencies
  }

  bool detectNotch(List<double> thresholds) {
    // Check for dip at 3-6 kHz with recovery at 8 kHz
  }

  // Model Inference
  AudiologicalAssessment analyze(AudiogramData data) {
    // 1. Extract features
    // 2. Run through TFLite model
    // 3. Apply post-processing rules
    // 4. Generate assessment
  }
}
```

Part 3: Training Data Requirements

3.1 Minimum Dataset Size

Dataset Size	Model Performance	Recommendation
500-1,000	Basic functionality, high variance	Not recommended for production
1,000-5,000	Moderate accuracy (85-90%)	Acceptable for MVP
5,000-10,000	Good accuracy (90-95%)	Recommended minimum
10,000+	Excellent accuracy (95%+)	Ideal for production

Research Reference: One study used 12,518 audiograms from 6,259 patients to achieve robust classification.

3.2 Data Labeling Requirements

Each audiogram must include expert labels for:

Label Category	Values	Labeler
Type of Loss	Normal, Conductive, Sensorineural, Mixed	Audiologist
Degree of Loss (per ear)	Normal, Mild, Moderate, Mod-Severe, Severe, Profound	Calculated from PTA
Configuration	Flat, Sloping, Rising, Cookie-bite, Notched, Precipitous	Audiologist
Symmetry	Symmetric, Asymmetric	Calculated
Additional Conditions	Otosclerosis, NIHL, Presbycusis, etc.	Audiologist

3.3 Data Schema


```
{
  "id": "audiogram_001",
  "patient_demographics": {
    "age": 45,
    "gender": "M",
    "occupation": "Factory Worker",
    "noise_exposure_history": true
  },
  "test_metadata": {
    "date": "2026-01-15",
    "audiometer": "GSI AudioStar Pro",
    "headphones": "TDH-39",
    "masking_used": true,
    "environment": "Sound Booth"
  },
  "pure_tone_thresholds": {
    "right_ear": {
      "air_conduction": {
        "250": 15, "500": 20, "1000": 25, "2000": 35,
        "3000": 50, "4000": 55, "6000": 45, "8000": 40
      },
      "bone_conduction": {
        "250": 15, "500": 20, "1000": 20, "2000": 30,
        "3000": 45, "4000": 50
      }
    },
    "left_ear": {
      "air_conduction": {
        "250": 10, "500": 15, "1000": 20, "2000": 30,
        "3000": 45, "4000": 50, "6000": 40, "8000": 35
      },
      "bone_conduction": {
        "250": 10, "500": 15, "1000": 20, "2000": 30,
        "3000": 40, "4000": 45
      }
    }
  },
  "speech_audiometry": {
```

```
    "right_ear": {
      "srt": 30,
      "wrs": 92,
      "wrs_presentation_level": 70
    },
    "left_ear": {
      "srt": 25,
      "wrs": 96,
      "wrs_presentation_level": 65
    }
  },
  "labels": {
    "type_of_loss": "sensorineural",
    "degree_right": "moderate",
    "degree_left": "mild",
    "configuration": "notched",
    "symmetry": "symmetric",
    "suspected_etiology": "noise_induced",
    "clinical_notes": "4kHz notch consistent with NIHL. Recommend hearing conservation."
  },
  "labeled_by": {
    "audiologist_id": "AUD_001",
    "credentials": "AuD, CCC-A",
    "date": "2026-01-15"
  }
}
```


3.4 Available Public Datasets

Dataset	Size	Content	Access
NHANES	9,000+ cases	Audiometric + demographic data	Public (USA)
biodatlab/autoaudiogram	200 audiograms	Annotated for object detection	GitHub (request access)
DTU Research Database	~28 listeners	Audiograms + population data	Research access
Purdue ARDC	Growing	Audiological + survey measures	Research collaboration

3.5 Data Augmentation Strategies

For limited datasets, augmentation can improve model robustness:

```
# For Image-Based Models
- Random rotation (±5°)
- Random horizontal flip
- Color jitter (for varied audiogram software)
- Random crop and resize
- Adding simulated noise

# For Numerical Data
- Add Gaussian noise to thresholds (±3-5 dB)
- Interpolate new samples between existing data points
- SMOTE for class balancing
- Create synthetic mixed hearing loss from conductive + SNHL combinations
```

Part 4: Rule-Based Enhancements

4.1 Deterministic Rules for Validation

While ML handles complex patterns, deterministic rules ensure safety:

```
class AudiologicalRules {

 // Rule 1: Air-Bone Gap Detection
 bool hasSignificantABG(AudiogramData data) {
   for (var freq in [500, 1000, 2000, 4000]) {
     double gap = data.ac[freq] - data.bc[freq];
     if (gap >= 15) return true;  // Significant conductive component
   }
   return false;
 }

 // Rule 2: Type Classification by ABG
 HearingLossType classifyType(AudiogramData data) {
   double avgBC = calculatePTA(data.bc);
   double avgAC = calculatePTA(data.ac);
   double avgABG = avgAC - avgBC;

   if (avgAC <= 25 && avgBC <= 25) {
     return HearingLossType.normal;
   } else if (avgABG >= 15 && avgBC <= 25) {
     return HearingLossType.conductive;
   } else if (avgABG < 10 && avgAC > 25) {
     return HearingLossType.sensorineural;
   } else if (avgABG >= 10 && avgBC > 25) {
     return HearingLossType.mixed;
   }
   return HearingLossType.undefined;
 }

 // Rule 3: Asymmetry Check (Red Flag)
 bool hasAsymmetry(AudiogramData right, AudiogramData left) {
   double ptaR = calculatePTA(right.ac);
   double ptaL = calculatePTA(left.ac);
   return (ptaR - ptaL).abs() >= 15;  // >15 dB difference
 }

 // Rule 4: SRT-PTA Correlation
 bool validateSRT(double srt, double pta) {
   // SRT should be within 5-12 dB of PTA
   return (srt - pta).abs() <= 12;
 }

 // Rule 5: NIHL Pattern Detection
 bool detectNIHLPattern(AudiogramData data) {
```

```
    // Check for notch at 3-6 kHz with recovery at 8 kHz
    double threshold4k = data.ac[4000];
    double threshold8k = data.ac[8000];
    double threshold2k = data.ac[2000];

    // Notch: 4kHz is at least 15 dB worse than 2kHz,
    // and 8kHz is at least 10 dB better than 4kHz
    return (threshold4k - threshold2k >= 15) && (threshold4k - threshold8k >= 10);
  }

  // Rule 6: Degree Classification (WHO/ASHA)
  HearingLossDegree classifyDegree(double pta) {
    if (pta <= 25) return HearingLossDegree.normal;
    if (pta <= 40) return HearingLossDegree.mild;
    if (pta <= 55) return HearingLossDegree.moderate;
    if (pta <= 70) return HearingLossDegree.moderatelySevere;
    if (pta <= 90) return HearingLossDegree.severe;
    return HearingLossDegree.profound;
  }

  // Rule 7: Clinical Red Flags
  List<ClinicalFlag> detectRedFlags(AudiogramData right, AudiogramData left) {
    List<ClinicalFlag> flags = [];

    if (hasAsymmetry(right, left)) {
      flags.add(ClinicalFlag.asymmetricLoss);  // Could indicate acoustic neuroma
    }

    if (detectSuddenOnset(right) || detectSuddenOnset(left)) {
      flags.add(ClinicalFlag.suddenOnset);  // REF to ENT urgently
    }

    if (lowWRS < 60) {
      flags.add(ClinicalFlag.poorSpeechDiscrimination);  // Possible retrocochlear
    }

    return flags;
  }
}
```

4.2 Hybrid AI + Rules Architecture

```
┌──────────────────────────────────────────┐
│              AI Audiologist Architecture │
├──────────────────────────────────────────┤
│                                          │
│  Raw Audiogram Data                      │
│       ↓                                  │
│  ┌────────────────────────────────────┐  │
│  │       Feature Extraction           │  │
│  │ • Threshold values (AC, BC per frequency per ear) │  │
│  │ • PTA calculations                 │  │
│  │ • ABG calculations                 │  │
│  │ • Slope/configuration features     │  │
│  └────────────────────────────────────┘  │
│       ↓                                  │
│  ┌───────────────┐  ┌───────────────┐    │
│  │ ML Model      │  │ Rule Engine   │    │
│  │ (TFLite/ONNX) │  │ (Deterministic) │  │
│  │               │  │               │    │
│  │ • Type pred.  │  │ • ABG rules   │    │
│  │ • Config pred.│  │ • Degree calc │    │
│  │ • Confidence  │  │ • Red flags   │    │
│  └───────┬───────┘  └───────┬───────┘    │
│          │                  │            │
│          └────────┬─────────┘            │
│                   ↓                      │
│  ┌────────────────────────────────────┐  │
│  │       Ensemble/Fusion Logic        │  │
│  │ • If ML confidence < 80%, defer to rules │  │
│  │ • If rules detect red flag, override ML │  │
│  │ • Combine predictions with confidence weighting │  │
│  └────────────────────────────────────┘  │
│       ↓                                  │
│  Final Assessment + Recommendations      │
│                                          │
└──────────────────────────────────────────┘
```

Part 5: FDA and Regulatory Considerations

5.1 FDA Precedents for AI Hearing Devices

Device/Software	Company	FDA Status	Date	Classification
Hearing Aid Feature (HAF)	Apple	De Novo Authorized	Sept 2024	Class II OTC
Tuned AI Hearing Assistant	Tuned	FDA Cleared	Oct 2025	Class II OTC
Sontro Self-Fitting	Soundwave	510(k) Cleared	2022	Class II OTC
Nuance Audio	Nuance	FDA Approved	2024	Class II OTC

5.2 Regulatory Pathway for AI Audiologist

Scenario A: Screening Only (Lower Regulatory Burden)

```
┌─────────────────────────────────────────────────────────────┐
│  "AI-Powered Hearing Wellness Check"                        │
│                                                             │
│  Claims Allowed:                                            │
│     "Identifies when to see a professional"                 │
│     "Tracks changes in hearing over time"                   │
│     "Visual audiogram with educational information"         │
│                                                             │
│  Claims NOT Allowed:                                        │
│     "Diagnoses hearing loss type"                           │
│     "Clinical-grade accuracy"                               │
│     "Replaces audiologist assessment"                       │
│                                                             │
│  Pathway: Enforcement Discretion / General Wellness         │
└─────────────────────────────────────────────────────────────┘
```

Scenario B: Clinical Use (FDA Submission Required)

```
┌──────────────────────────────────────────────────────────
──────────────────────────────┐
│  "AI-Powered Audiometric Interpretation Software"            │
│                                                          │
│  Claims Allowed (after approval):                           │
│      "Classifies hearing loss type with X% accuracy"           │
│      "Automated interpretation for clinical workflow"          │
│      "Assists audiologists with routine interpretation"        │
│                                                          │
│  Regulatory Requirements:                                   │
│  • IEC 62304 Software Lifecycle Documentation               │
│  • Clinical Validation Study (n≥100)                        │
│  • Algorithm Performance Report                             │
│  • 510(k) or De Novo Submission                             │
│  • Post-market surveillance plan                            │
│                                                          │
│  Pathway: 510(k) or De Novo (Class II)                      │
│  Timeline: 12-24 months                                     │
│  Cost: $50,000-$150,000                                     │
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```

5.3 Labeling Requirements for AI Features

Required disclaimers for AI-based interpretation:

> **For Patients:**
> "This AI-powered assessment is designed to provide educational information about your hearing health. It is not a medical diagnosis. If you have concerns about your hearing, please consult a licensed audiologist or ENT physician."

> **For Professionals:**
> "This AI interpretation is intended to assist clinical decision-making and should be reviewed by a qualified audiologist. The AI may not account for all clinical factors. Final interpretation and treatment decisions remain with the healthcare provider."

Part 6: Implementation Roadmap

6.1 Data Collection Phase (Months 1-3)

Task	Duration	Resources Needed
Partner with 3-5 audiology clinics	2 weeks	Business development
Develop data collection protocol	2 weeks	Data scientist + audiologist
Build data annotation tool	3 weeks	Flutter developer
Collect 1,000+ labeled audiograms	8 weeks	Partner audiologists
Clean and validate dataset	2 weeks	Data scientist

Cost Estimate: $15,000 - $25,000

6.2 Model Development Phase (Months 3-5)

Task	Duration	Resources Needed
Feature engineering	2 weeks	ML engineer
Train baseline models (RFC, XGBoost)	2 weeks	ML engineer
Train deep learning models	3 weeks	ML engineer + GPU resources
Hyperparameter tuning	2 weeks	ML engineer
Model evaluation and selection	1 week	ML engineer + audiologist
Convert to TFLite/ONNX	1 week	ML engineer

Cost Estimate: $25,000 - $40,000 (includes cloud compute)

6.3 Integration Phase (Months 5-6)

Task	Duration	Resources Needed
Integrate TFLite model into Flutter app	3 weeks	Flutter developer
Build rule engine overlay	2 weeks	Flutter developer
Implement feedback collection	1 week	Flutter developer
UI/UX for AI results display	2 weeks	Designer + developer

Cost Estimate: $10,000 - $15,000

6.4 Validation Phase (Months 6-8)

Task	Duration	Resources Needed
Clinical validation study (n=100+)	6 weeks	Partner audiologists
Statistical analysis and reporting	2 weeks	Data scientist
Model refinement based on results	2 weeks	ML engineer

Cost Estimate: $15,000 - $30,000

6.5 Total AI Development Investment

Phase	Minimum	Maximum
Data Collection	$15,000	$25,000
Model Development	$25,000	$40,000
Integration	$10,000	$15,000
Validation	$15,000	$30,000
Total	**$65,000**	**$110,000**

Part 7: Limitations and Challenges

7.1 What AI Cannot Replace

Task	AI Capability	Reason
Physical examination (otoscopy)	Not possible	Requires camera/scope

Tympanometry/immittance	Not possible	Requires specialized hardware
OAE testing	Not possible	Requires specialized probe
ABR testing	Not possible	Requires electrodes + equipment
Complex case counseling	⚠ Limited	Requires empathy, context
Hearing aid fitting	⚠ Partial	Apple/Tuned doing basic fitting
Legal liability for diagnosis	Not possible	Licensed professional required

7.2 Technical Challenges

Challenge	Impact	Mitigation
Data quality variability	Model performance degradation	Strict data collection protocols
Dataset bias	Poor performance on underrepresented groups	Diverse data collection, augmentation
Mixed hearing loss classification	Lower accuracy (hardest category)	Focus on binary: conductive component Y/N
Consumer headphone calibration	Threshold accuracy varies	Implement reference tone validation
Edge cases and rare conditions	Misclassification risk	Strong "refer to specialist" rules

7.3 Recommended Scope Limitations

AI Can Do:

- Classify hearing loss type (conductive/sensorineural/mixed)
- Calculate degree of hearing loss
- Identify audiogram configuration patterns
- Detect NIHL notch patterns
- Flag asymmetric hearing loss
- Generate educational reports
- Recommend "see a professional" when indicated

AI Should NOT Do:

- Definitively diagnose specific conditions (e.g., otosclerosis, Ménière's)
- Recommend specific hearing aids
- Provide treatment advice
- Replace professional consultation for any flagged cases
- Make autonomous decisions about medical referrals

Part 8: Recommendations

8.1 Recommended Approach

Based on this research, we recommend a **phased approach**:

Phase 1: Rule-Based System (Immediate)

- Implement deterministic rules for type, degree, configuration
- No ML model needed initially

- Can deploy immediately
- Low regulatory burden

Phase 2: ML Enhancement (Months 4-8)

- Collect real-world data from app usage
- Train models on collected data
- Deploy as "AI-assisted" rather than "AI-autonomous"
- Keep human oversight for complex cases

Phase 3: Full AI (Months 12-24)

- After sufficient validation data
- FDA submission if making clinical claims
- Consider "AI Audiologist" branding only after regulatory clearance

8.2 Competitive Advantage

While not fully replacing audiologists, AI can provide:

1. **Instant screening results** - No waiting for appointments
2. **Consistent interpretation** - Reduces human variability
3. **24/7 availability** - Remote/rural access
4. **Cost reduction** - Lower per-test cost
5. **Scalability** - Handle millions of tests simultaneously
6. **Longitudinal tracking** - AI can track changes over time

8.3 Recommended Next Steps

1. **Decide on regulatory pathway** - Wellness vs. medical device
2. **Begin data collection partnerships** - Partner with 3-5 clinics
3. **Implement rule-based system** - Start with deterministic approach
4. **Build data collection infrastructure** - Within the app
5. **Consult FDA (Q-Sub)** - If pursuing clinical claims
6. **Hire/contract ML engineer** - For model development

Summary: Feasibility Assessment

Question	Answer
Is AI audiometry interpretation technically feasible?	**Yes** - 97%+ accuracy demonstrated
Can it fully replace audiologists?	**No** - But can handle routine interpretation
Is there FDA precedent?	**Yes** - Apple, Tuned approved in 2024-2025
What data is needed?	5,000+ labeled audiograms (AC, BC, speech, labels)
What is the cost?	$65,000 - $110,000 for AI development
What is the timeline?	6-12 months for functional AI
Should we proceed?	**Yes, with phased approach**

References

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5. ANSI S3.6-2018 - Specification for Audiometers
6. ISO 8253-1:2010 - Audiometric Test Methods
7. Computational Audiology Community Resources